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The virulence function of RXLR effector *PpE4* of *Phytophthora parasitica* is coupled with its cell death-inducing activity

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ABSTRACT

RXLR effectors secreted by *Phytophthora* pathogens are important virulence factors in suppressing plant immunity and facilitating pathogen infection. In addition to possessing a variety of functions, some RXLR effectors have been found to trigger host cell death with different underlying mechanisms. *PpE4* was previously reported as a virulent RXLR effector secreted by *Phytophthora parasitica* and triggers cell death in various plants. However, it is uncertain whether this type of cell death promotes or restricts the colonization of this hemibiotrophic pathogen. Here, we explored the correlation between the cell death activity of *PpE4* and its contribution to the infection of *P. parasitica*. Sequencing data revealed that *PpE4* exhibited both point mutations and presence/absence polymorphisms in the *P. parasitica* population under positive selection. *E4D*, one of four alleles, lost its cell death-inducing ability, also lost its ability to increase plant susceptibility to pathogen infection. Furthermore, *E4D* could compromise the cell death-inducing and virulence functions of the other three *PpE4* alleles. *PpE4* truncated mutants deficient in cell death-inducing activity showed abolished ability to increase plant susceptibility. Finally, *PpE4* was unable to increase the susceptibility of TRV-NPK plants, which showed abolished cell death. Taken together, the cell death-inducing activity of *PpE4* is essential for its virulence function during colonization of hemibiotrophic pathogen, suggesting that *P. parasitica* exploits the cell death triggered by *PpE4* to facilitate the transition from biotrophy to necrotrophy. These findings provide a new perspective for understanding the role of RXLR effectors triggered cell death in the pathogenesis of hemibiotrophic pathogens.

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Introduction

Phytophthora parasitica (Synonyms: *P. nicotianae* [1]) is a model oomycete pathogen with a wide range of host plants [2]. It causes devastating diseases of agricultural and horticultural crops, such as tobacco black shank and citrus root rot and gummosis [3]. *P. parasitica*, along with many pathogens from the *Phytophthora* genus, are hemibiotrophic pathogens, they initiate infection in living plants and maintain a biotrophic lifestyle in the early infection stage, then switch to a necrotrophic lifestyle later [4]. Cell death at the early infection stage can restrict pathogen colonization, while cell death occurred at the later infection stage is conducive to the transition from biotrophy to necrotrophy for hemibiotrophic pathogens [5]. Therefore, the regulation of host cell death is particularly important in the interaction between plants and hemibiotrophic pathogens [6].

RXLR effectors are oomycete pathogen-derived small molecular proteins containing a secretion signal peptide followed by the Arg-X-Leu-Arg (RXLR) motif [7]. Certain RXLR effectors can suppress pathogen-associated molecular patterns-triggered immunity (PTI) or effector-triggered immunity (ETI) and help pathogens break through two layers of plant immunity [8–10]. Additionally, some effectors interact with a variety of host targets and disrupt normal plant cellular processes, thereby creating a favorable environment to support pathogen colonization [11–14]. Their virulence function is determined by their ability to manipulate host defenses and promote pathogen infection; thus, the pathogenicity of pathogens would be affected when these effector genes are silenced or knocked out [15–17]. During the coevolution of plants and phytopathogens, specific RXLR effectors have been recognized by plant intracellular nucleotide-binding, leucine-rich repeat receptors (NLRs) to trigger ETI [18–21]. ETI are usually accompanied by

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plant cell death called the hypersensitive response (HR), which is a programmed cell death controlled by plant [22]. Effectors that trigger NLRs-mediated cell death in plants, such as *P. infestans* Avrblb2 and *P. sojae* Avr1d [23,24] have been defined as avirulence (AVR) effectors.

Studies have found that many RXLR effectors, such as Avh241 and Avh238 from *P. sojae* [25,26], Pi23226 from *P. infestans* [27], PcAvh1 from *P. capsici* [28], PvAvh74 from *Plasmopara viticola* [29], and PIAvh142 from *Peronophythora litchi* [30], can induce nonspecific cell death in plants, which is distinct from the cell death triggered by AVRs. Unlike the HR mediated by host-specific receptor proteins, this form of cell death is potentially mediated by unidentified nonspecific receptors. Furthermore, the cell death induced by effectors such as PcAvh1, PIAvh142 and Vd424Y does not restrict pathogen expansion. Instead, these effectors are essential for the virulence of the pathogens and promote infection when transiently expressed in plants [28,30,31]. Additionally, the *P. capsici* virulent RXLR effector CRISIS2 associates with the plasma membrane H^+ -ATPase (PMA) to trigger plant cell death in an immune receptor-independent manner [32]. Similarly, the *P. infestans* RXLR effector PITG06478 triggers cell death by interacting with 14-3-3 to inhibit PMA activity [33]. Another type of effector, such as PsCRN63, also promotes *P. capsici* infection while inducing cell death in plant by interacting with catalases [34]. Another interesting example is the necrotrophic fungus *Cochliobolus victoriae*, which produces an effector, called victorin. This effector activates an NLR-mediated cell death to facilitate *C. victoriae*'s necrotrophy [35]. Therefore, effector-induced cell death is not always a result of plant defense response, but can also be a strategy of hemibiotrophic or necrotrophic pathogens to promote their colonization.

Previous studies have revealed that PpE4, an RXLR effector secreted by *P. parasitica* during infection, is capable of inducing nonspecific cell death in a variety of plants, including tobacco, tomato, potato and Arabidopsis. Moreover, silencing of *PpE4* impacts the pathogenicity of *P. parasitica*, while overexpression of *PpE4* in plants promotes *P. parasitica* infection [36]. The cell death induced by *PpE4* is nonspecific and relies on components of the ETI/PTI pathways, implying that the recognition receptors or mechanisms for triggering cell death are conserved across different plant species. However, as a virulence effector, the correlation between its triggered cell death and its virulence function is still not fully understood. It remains unclear whether the cell death induced by *PpE4* limits *P. parasitica* invasion or contributes to pathogen colonization. In the present study, we investigated the polymorphism of *PpE4*

in *P. parasitica* populations and identified an allele lacking both cell death-inducing ability and virulence function. Surprisingly, this allele was able to inhibit the cell death-inducing and virulence functions of the other three alleles. To further elucidate the relationship between cell death and virulence, we assayed the impact of *PpE4* on infection when its cell death was blocked. Our data demonstrated that the cell death-inducing activity of *PpE4* is coupled with its ability to enhance plant susceptibility, which implies that *P. parasitica* takes advantage of the cell death induced by *PpE4* to aid in the switch from biotrophy to necrotrophy, rather than restricting pathogen invasion.

Materials and methods

Materials and plant growth conditions

These experimental materials including bacterial strains, *P. parasitica* strains, tobacco seeds, and *Nicotiana benthamiana* seeds are reserved in State Key Laboratory for Crop Stress Resistance and High-Efficiency Production., *N. benthamiana* and tobacco seeds were routinely cultured in a matrix in a climate chamber for about 5–6 weeks in a 22–25 °C climate chamber with a photoperiod of 14 h light, 10 h dark and 70% relative humidity.

Vector construction and polymorphism analysis

All the primers and vectors used in this study are listed in Table S1. The plasmids and PCR products were digested by appropriate restriction endonucleases (Promega, Madison, WI, USA) followed by ligation using T4 DNA ligase (Promega, Madison, WI, USA). For the polymorphism assay, *PpE4* coding region-specific primers (E4FL-F/E4FL-R) and primers (E4L-F/E4L-R) specifically binding to the upstream and downstream regions of the *PpE4* ORF were designed, according to the genomes sequencing data of different *P. parasitica* strains. The *PpE4* coding region (405 bp) and the *PpE4* locus (596 bp) were amplified and cloned into the pMD19-T vector (Takara, Dalian, China) to be sequenced. The sequences of point mutants were synthesized in AZENTA. All constructs were sequenced in Genscript (Nanjing, China). Protein sequences of *PpE4* alleles were aligned by ClustalW (Version 2.1) algorithm at default parameters in the MegAlign module.

Agrobacterium tumefaciens injection in *N. benthamiana*

Plant binary expression vectors were transformed into *A. tumefaciens* GV3101. Monoclonal colonies were cultured in LB medium with the appropriate antibiotics at

28°C for 2 days, then bacteria were harvested and resuspended in MES buffer with 200 μ M acetosyringone as previously described [36]. For cell death test, *A. tumefaciens* were infiltrated into the *N. benthamiana* leaves at a final optical density (OD₆₀₀) of 0.4, and for *P. parasitica* inoculation, infiltrations were performed at OD₆₀₀ of 0.01. VIGS assay was done as previously described [36], *A. tumefaciens* GV3101 carrying pTRV1 and pTRV2:*NPK*, respectively, were mixed in equal amounts to a final concentration of OD₆₀₀ of 0.25, pTRV2:*GFP* was used as control. Injection was performed on the two lower leaves of *N. benthamiana* at the 4-leaf stage.

After 16–20 days, gene silencing efficiency of the upper leaves was analyzed, then *PpE4* and *GFP* was transiently expressed on the silenced leaves followed by *P. parasitica* inoculation.

Electrolyte leakage assay

Cell death was assayed by measuring ion leakage from leaf discs [26]. Briefly, for each sample, five leaf discs (diameter 1 cm) were floated on 5 ml of distilled water for 3 h at room temperature. Then, the conductivity of the bathing solution was measured with a conductivity meter (S470 Seven Excellence, Mettler Toledo, Shanghai, China) to yield “value A.” The leaf discs were then returned to the bathing solution and boiled in sealed tubes for 25 min. After cooling the solution to room temperature, the conductivity was measured again to obtain “value B.” Ion leakage was expressed as the percentage of total ions, that is (value A/value B)×100%. All assays were repeated three times.

Inoculation of *P. parasitica*

P. parasitica strain 1121, a transformant line of strain Pp016 stably expressing free GFP, was used in the inoculation assay [35]. Strain 1121 was verified through antibiotic selection to verify its stability and cultured on 5% CA solid medium at 23°C for 4 days, then the mycelium plugs were transferred to 5% CA liquid medium for another 4 days. The 5% CA liquid medium was replaced with Petri solution to induce sporangia production. Finally, the culture was treated with pre-cooled sterilized distilled water for 30 minutes to release zoospores as described previously [36]. *N. benthamiana* and tobacco seeds were cultured in a matrix containing soil and vermiculite in a 22–25°C climate chamber with a photoperiod of 14 h light, 10 h dark and 70% relative humidity for about 5 to 6 weeks. About 10³ *P. parasitica* 1121 zoospores were inoculated on each side of the leaves one day post *A. tumefaciens* infiltration. The

diameters of water-soaked lesion were measured at 36–48 hours post-inoculation (hpi). The biomass was calculated by the DNA ratio of *P. parasitica* to *N. benthamiana* on leaf of identical areas using qPCR with *PpUBC*-specific primers and *N. benthamiana* *F_Box*-specific primers (Table S1). The expansion of *P. parasitica* hyphae was stained by trypan blue.

Gene expression analyses by RT-qPCR

For the expression analysis of *PpE4* during infection process, hyphae of different *P. parasitica* strains from 4-day-old 5% CA liquid medium and *N. benthamiana* leaves inoculated with mycelium plugs at 24, 48 and 72 hpi were harvested for RNA extraction using TRIzol reagent (Invitrogen, Carlsbad, CA, USA) followed by reverse transcription using a PrimeScript™ RT Reagent Kit with gDNA Eraser (TaKaRa, Dalian, China) according to the product manuals. Reverse transcription-quantitative polymerase chain reaction (RT-qPCR) was performed using SYBR Green mix (CWBio, Beijing, China) on a QuantStudio™ 3 Real-Time PCR System (Thermo Scientific, Waltham, MA, USA). Ubiquitin-conjugating enzyme (*PpUBC*) and 40S ribosomal protein S3A (*PpWS21*) genes of *P. parasitica* [37] and the β -actin gene of *N. benthamiana* were used as internal controls in RT-qPCR.

Western blot analysis

The protein extraction and western blot were performed as described previously [36]. In brief, proteins of *N. benthamiana* leaves were extracted at 2 dpi using RIPA lysis buffer with 10 mM dithiothreitol (DTT) and separated by SDS-PAGE. Then the proteins were transferred to PVDF membrane (Roche, Basel, Switzerland) and blocked in 10% skimmed milk for 2 h. Next, the PVDF membrane was incubated in mouse anti-Flag antibody (Abbkine, Redlands, CA, USA) or mouse anti-mCherry antibody (Abbkine, Redlands, CA, USA) solution with 1:2000 dilution at room temperature for 1.5–2 h, followed by washing and incubating in goat anti-mouse antibody for 1.5–2 h. Finally, the protein bands were visualized under a ChemiDOC™ XRS+ imaging system (Bio-Rad Laboratories, Hercules, CA) using an eECL Western blot kit (CWBio, Beijing, China).

Co-immunoprecipitation (Co-IP) assay

E4A-GFP and free GFP, respectively, were co-expressed with Flag-E4D in *N. benthamiana* leaves by agroinfiltration. Total proteins were extracted from the *N. benthamiana*

leaves using RIPA buffer containing 1 mM DTT. Subsequently, the proteins were incubated with ChromoTek GFP-Trap agarose (Proteintech, Wuhan, China) to capture the GFP-tagged proteins along with their potential interacting proteins according to the product manual. Finally, the proteins in both input and output solutions were detected by western blot using anti-GFP (Proteintech, Wuhan, China) and anti-Flag (Abbkine, Redlands, CA, USA) antibodies, respectively.

Results

PpE4 exhibits substantial expression and sequence polymorphisms in *P. parasitica* isolates

PpE4 is highly expressed during *P. parasitica* infection and triggers cell death in various plants [36]. To analyze

transcriptional polymorphism of *PpE4* in *P. parasitica* strains with different virulence, the expression patterns of *PpE4* during infection process were assayed by reverse transcription-quantitative polymerase chain reaction (RT-qPCR). *PpE4* was up-regulated at 24 hours post-inoculation (hpi) and expressed at low levels in mycelium and late infection stages in two virulent strains (Pp014 and Pp016) (Figure 1(a)) [38], which is consistent with previous results. However, its expression was not detectable in the less-virulent strains (Pp005 and Pp009) and undetermined strains (PpAY074 and PpAQ054) (Figure 1(a)). To explore the sequence polymorphism of *PpE4* in different *P. parasitica* isolates, the coding region of *PpE4* and its flanking region were amplified by PCR using genomic DNA from a collection of isolates of worldwide origin. Strikingly, no specific PCR products were

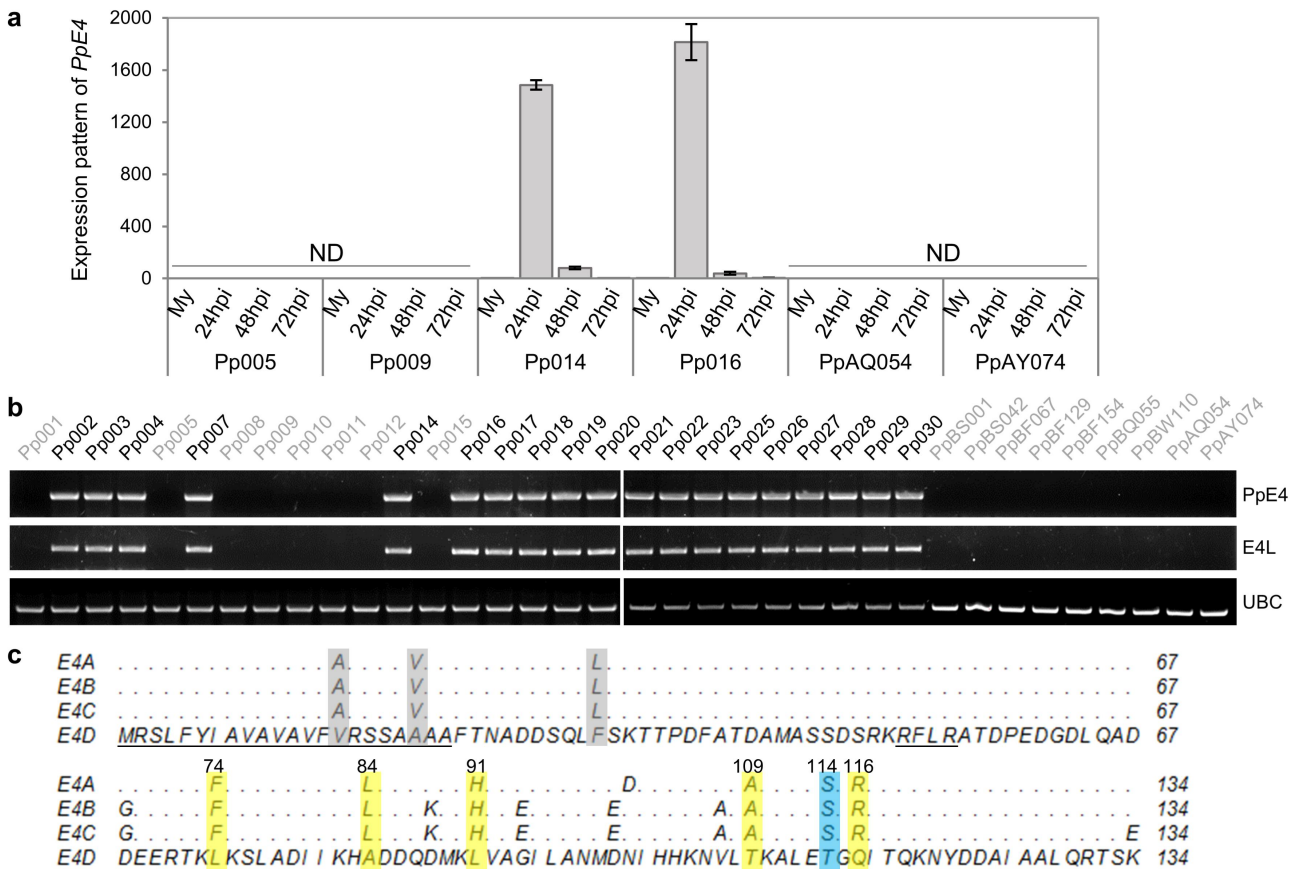


Figure 1. Point mutations and presence/absence polymorphisms of *PpE4* in *Phytophthora parasitica* population. (a) The transcript of *PpE4* is completely undetectable in some strains. Reverse transcription-quantitative PCR (RT-qPCR) was used to quantitate relative expression level of *PpE4* in different stages of six strains. Pp005 and Pp009, less-virulent strains; Pp014 and Pp016, virulent strains; PpAY074 and PpAQ054, strains collected from Chongqing with undetermined virulence. My, mycelia grown in 5% CA medium; infected *Nicotiana benthamiana* leaves harvested at 24, 48 and 72 hours post-inoculation (hpi) of *P. parasitica* mycelia plugs. *PpUBC* is used as the reference gene, and the Fold change of *PpE4* was normalized to the expression in Pp016 mycelia. Error bars indicate standard deviation (sd) of three biological replicates. ND, not detected, a signal below the detection limit threshold. (b) PCR amplification of *PpE4* (405 bp) and the *PpE4* locus (E4L, 596 bp) in 36 isolates from a worldwide collection. *PpUBC* was used as genome DNA control. The isolates with no specific bands detected were in grey, otherwise were in black. (c) Protein sequences alignment of *PpE4* alleles. Residues identical to E4D are represented by dots while substitutions are represented by the substituted residues. The signal peptide and RXLR motif are underlined. The N-terminal substitution sites not involved in cell death inducing are in gray. The C-terminal substitution sites with similar properties are marked in blue, while with different properties are in yellow.

obtained in 17 out of the 36 strains using either *PpE4*-specific primers or the flanking region-specific primers (Figure 1(b)). No transcripts were detected, and no fragments were amplified in the genomic DNA, suggesting that the corresponding *PpE4* locus is highly divergent or may even be absent in the genomes of these isolates. Further analysis of the draft genomes of 14 sequenced *P. parasitica* strains revealed that the *PpE4* locus was absent in the genomes of seven strains (Table 1). Moreover, *PpE4* was not included in the 93 “core RXLR effectors” predicted to exist in nine sequenced strains [39], which further confirmed that the *PpE4* locus is absent in the genomes of numerous strains.

To further investigate the sequence variation, *PpE4* sequences from 19 strains were amplified and inserted into vector pMD19-T for sequencing. Based on the sequencing results and genome data, four different *PpE4* variants were identified, each encoding a distinct protein: E4A (previously known as PpE4), E4B, E4C and E4D (Figure 1(c) and Table 1). A total of 23 single nucleotide polymorphism (SNP) sites were detected in four alleles, resulting in the substitution of 16 amino acids, 13 of which occur at the C-terminus (Figure 1(c)). The ratio of non-synonymous nucleotide substitutions to synonymous nucleotide substitutions (dN/dS) at the C-terminus is 1.14, as calculated by MEGA7 software [40], indicating that *PpE4* may have

Table 1. *PpE4* alleles in different *Phytophthora parasitica* isolates.

Strains	<i>PpE4</i> alleles ^a	Host plant	Region	Mating type
Pp001	N	—	—	A2
Pp002	C	Eggplant	Western Australia	A2
Pp003	C	Eggplant	—	A2
Pp004	A	Tobacco	—	A1
Pp005	N	<i>Nicotiana glauca</i>	Victoria Australia	A2
Pp007	A	Tobacco	Queensland Australia	A1
Pp008	N	Tobacco	Queensland Australia	A2
Pp009	N	Tobacco	Queensland Australia	A2
Pp010	N	Pawpaw	Queensland Australia	A2
Pp011	N	Pawpaw	Queensland Australia	A2
Pp012	N	Pawpaw	Queensland Australia	A2
Pp014	A	Tobacco	Queensland Australia	A2
Pp015	C	Tobacco	Queensland Australia	A2
Pp016	A	Tobacco	Queensland Australia	A1
Pp017	B	—	—	A2
Pp018	B	<i>Banksia</i> species	Western Australia	A2
Pp019	B	—	Western Australia	A2
Pp020	B	<i>Banksia</i> species	Western Australia	A2
Pp021	B	<i>Citrus</i> species	—	A1
Pp022	B	<i>Citrus</i> species	—	A1
Pp023	B	<i>Citrus</i> species	—	A1
Pp025	C	<i>Dendrobium candidum</i>	Zhejiang China	A2
Pp026	C	<i>Dendrobium candidum</i>	Zhejiang China	A2
Pp027	C	<i>Dendrobium candidum</i>	Zhejiang China	A2
Pp028	C	<i>Dendrobium candidum</i>	Zhejiang China	A2
Pp029	C	<i>Dendrobium candidum</i>	Zhejiang China	A2
Pp030	C	<i>Dendrobium candidum</i>	Zhejiang China	A2
PpBS001	N	Tobacco	Chongqing China	A2
PpBS042	N	Tobacco	Chongqing China	A2
PpBF067	N	Tobacco	Chongqing China	A2
PpBF129	N	Tobacco	Chongqing China	A2
PpBF154	N	Tobacco	Chongqing China	A2
PpBQ055	N	Tobacco	Chongqing China	A2
PpBW110	N	Tobacco	Chongqing China	A2
PpAQ054	N	Tobacco	Chongqing China	A2
PpAY074	N	Tobacco	Chongqing China	A2
P10297.1	D	<i>Dieffenbachia exotica</i>	—	—
P149_1	D	—	—	—
INRA310.3	A	Tobacco	—	—
CHvinca01.1	B	Periwinkle	—	—
IAC_01_95	B	<i>Citrus</i> species	—	—
P1569_1	B	<i>Citrus</i> species	—	—
H02_1	B	—	—	—
P329	ND	—	—	—
P364	ND	—	—	—
PN47	ND	—	—	—
CJ01A1.1	ND	Tobacco	—	—
CJ02B3.1	ND	Tobacco	—	—
CJ05E6.1	ND	Tobacco	—	—
P1976.1	ND	Tomato	—	—

^a“N” means no *PpE4*-specific PCR product was amplified; “ND” means no *PpE4* sequence was detected in the genome data; “A,” “B,” “C” and “D” represent four different alleles, E4A, E4B, E4C and E4D, respectively. “—” means unknown information.

undergone positive selection. In addition, *E4A* was exclusively found in strains isolated from tobacco plants, while *E4B* and *E4C* were found in strains isolated from a variety of plants including tobacco, eggplants, citrus, *Dendrobium candidum*, and others. On the other hand, *E4D* was exclusively found in strains isolated from *Dieffenbachia maculate* and an unknown host. Multiple variants, including *E4A*, *E4C*, and deletion mutation, were found in tobacco isolates from different regions of Australia, whereas *PpE4* was not detected in strains isolated from pawpaw or tobacco from Chongqing (Figure 1 and Table 1). In conclusion, *PpE4* in the *P. parasitica* isolates exhibited both point mutations and presence/absence polymorphisms under positive selection. This genetic variation of *PpE4* is likely associated with host and regional adaptations.

All *PpE4* alleles induce cell death except for *E4D*

PpE4 was shown to induce cell death in various plant species [36]. To determine if all *PpE4* alleles possess this ability, *E4A*, *E4B*, *E4C*, and *E4D* were transiently expressed in the leaves of *N. benthamiana*, *N. glutinosa*, and two varieties of *N. tabacum*. All alleles, except *E4D*, were able to induce cell death, even though proteins from all alleles were accumulated in *N. benthamiana* leaves (Figure 2). There are nine amino acid residue differences between *E4D* and the other three alleles, with six located in the 56–121 region (Figure 1(c)). This particular region of *E4A* (*E4M8*) was demonstrated to be sufficient to trigger cell death [36]. To characterize key residues essential for the cell death-inducing activity of *PpE4*, we exchanged the different amino acids between *E4M8* and *DM8* (56–121 region of *E4D*) (Figure 3(a) and Figure S1). Following transient expression in *N. benthamiana* leaves,

proteins of the point mutants were expressed, however, the cell death symptoms and electrolyte leakage rate were significantly attenuated for M1, M2, and M4, while the cell death induced by M3, M5, and M6 was almost unaffected (Figure 3(b-d)). The mutations L84A, H91L, A109T, and R116Q had little impact on cell death induction of *E4M8*, whereas the F74L mutation lost its ability to induce cell death. This highlights that F74 of *PpE4* is essential for its cell death-inducing activity.

Cell death induced by *E4A*, *E4B*, and *E4C* could be suppressed by *E4D*

To determine whether *E4D*, the allele that does not trigger cell death, could suppress cell death induced by the other three alleles, each of the three alleles was transiently co-expressed with *E4D* or *GFP* in *N. benthamiana* (Figure 4 and Figure S2). Upon transient co-expression of each of the three alleles with *GFP* in *N. benthamiana*, rapid cell death was observed three days post agroinfiltration. Conversely, cell death of the three alleles was almost entirely abolished at the sites co-expressed with either mCherry-*E4D* or Flag-*E4D* (Figure 4(a) and Figure S2). Protein accumulation of mCherry-*E4D* and Flag-*E4A/B/C* was detected by anti-Flag antibodies and anti-mCherry antibodies by western blot (Figure 4(c)). The statistical analysis of the cell death scores revealed that *E4D* effectively counteracted cell death triggered by the other three alleles (Figure 4(b)). To investigate whether *E4D* could inhibit cell death triggered by *INF1*, an elicitor from *P. infestans* [41], *INF1* was transiently co-expressed with *E4D* or *GFP* in *N. benthamiana*. The results showed that *INF1* induced cell death rapidly when co-expressed with *GFP*, but when co-expressed with *E4D*, *INF1*-induced cell death was significantly weakened (Figure 4(d-e)). In addition, RXLR effectors may act

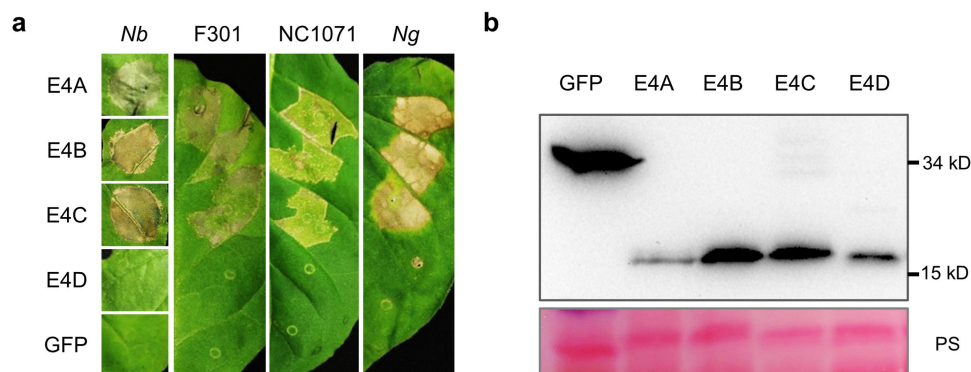


Figure 2. All alleles except *E4D* are able to induce cell death. (a) *Agrobacterium tumefaciens* carrying *PpE4* alleles were infiltrated in leaves of *Nicotiana benthamiana* (Nb), *N. glutinosa* (Ng) and two *N. tabacum* varieties (NC1071 and F301) respectively. Photographs were taken at 5 DPI. GFP, green fluorescent protein. (b) Proteins accumulation of *PpE4* alleles expressed in *N. benthamiana* were detected by western blot using anti-Flag antibodies. Protein loading is indicated by Ponceau S stain (PS). Three independent experiments were performed with similar results.

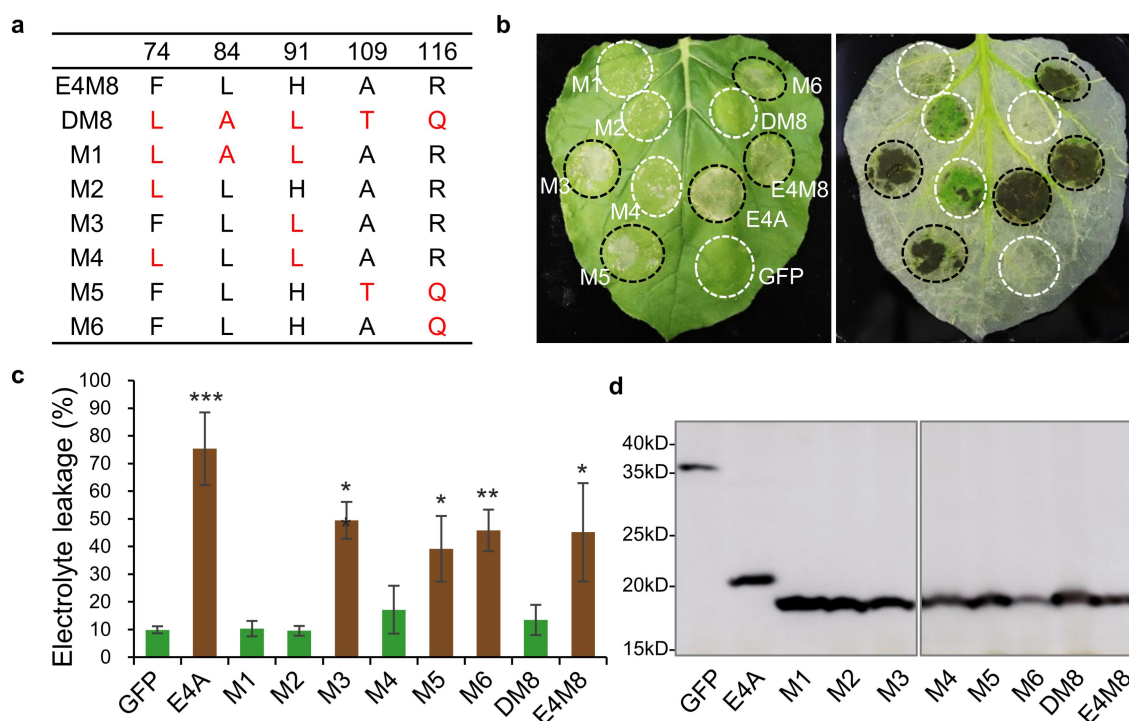


Figure 3. Phenylalanine at position 74 is crucial for the cell death-inducing activity of *PpE4*. (a) The design of the amino acid point mutants. (b) The cell death symptoms on *Nicotiana benthamiana* leaves expressing *PpE4* mutants. The leaf on the right is bleached with ethanol. (c) Quantification of cell death by electrolyte leakage measurement. Electrolyte leakage rate of *N. benthamiana* leaves expression *PpE4* mutants were measured as a percentage of leakage from boiled leaves. Error bars represent the standard deviation of three biological replicates. Asterisks denote significant differences from GFP control (one-sided Student's *t*-test: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$). (d) Protein accumulation of point mutants was detected by western blot using anti-Flag antibodies.

coordinated to effectively suppress host defenses. For instance, *Avh172*, *Avh240* and *Avh277* have been found to suppress cell death induced by *Avh238* from *P. sojae* [9]. Since *PpE4* and *Avh238* are orthologs [42], we examined whether *Avh172*, *Avh240* and *Avh277* could also suppress *E4A*-induced cell death. When co-expressed with *E4A* in *N. benthamiana*, *Avh172*, *Avh240* and *Avh277* were unable to inhibit cell death triggered by *E4A* (Figure S3).

All alleles of *PpE4* promote *P. parasitica* infection except for *E4D*

PpE4 (*E4A*) was previously found to induce cell death as well as promote *P. parasitica* colonization when expressed *in planta* [36]. To further investigate the impact of two types of *PpE4* alleles (which either trigger or suppress cell death) on pathogen infection, *E4B*, *E4C*, *E4D*, and a *GFP* control were transiently expressed in *N. benthamiana* leaves ($OD_{600} = 0.01$) one day before inoculation of *P. parasitica* zoospores. Measurement of lesion diameters and *P. parasitica* biomass showed that the expression of *E4B* and *E4C* enhanced *P. parasitica* colonization compared to the expression of *GFP*, while the expression of *E4D* inhibited pathogen colonization (Figure 5). These

results suggest that the allele losing cell death-inducing activity also loses the ability to enhance plant susceptibility to *P. parasitica* infection, even though it can inhibit the cell death induced by the other three alleles.

PpE4-mediated virulence contribution and cell death is suppressed by *E4D*

To figure out the correlation between *PpE4*-induced cell death and its virulence function, we investigated whether the cell death-inducing alleles of *PpE4* could still promote infection even when their cell death was suppressed by *E4D*. *E4A*, *E4B*, and *E4C* were co-expressed with *E4D* on the left sides of *N. benthamiana* leaves, respectively, while *GFP* was expressed on the right sides one day before inoculation of *P. parasitica* zoospores. There was no significant difference in the colonization of *P. parasitica* on each side of *N. benthamiana* leaves (Figure 6), suggesting that *E4D* was able to neutralize the contribution of the other three alleles to pathogen infection by suppressing their cell death. Co-IP assay was performed to explore whether this inhibition occur through direct interaction between *E4D* and the other alleles. The

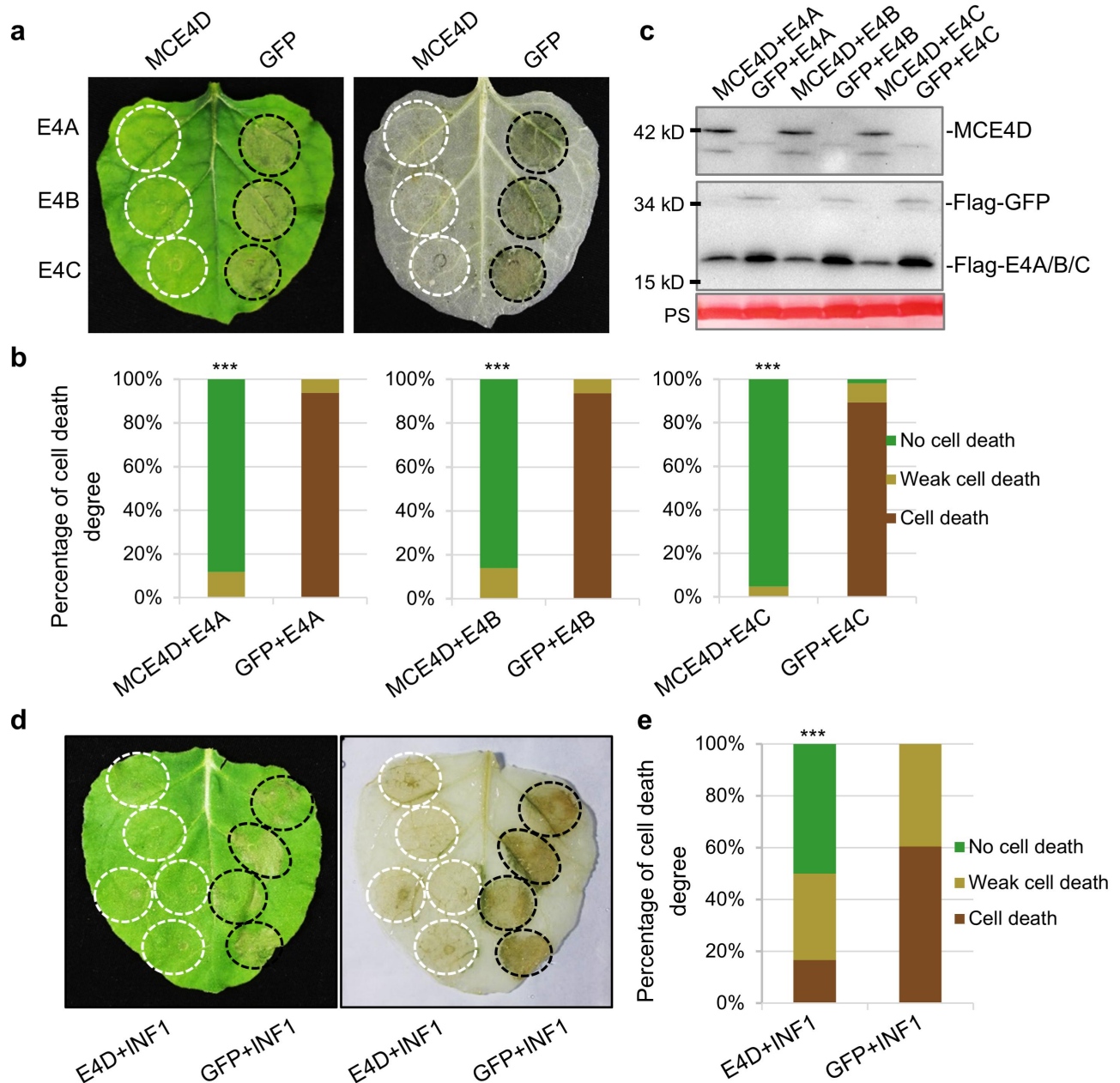


Figure 4. *E4D* can suppress the cell death induced by *E4A*, *E4B*, *E4C* and *INF1*. *Agrobacterium tumefaciens* carrying *MCE4D* and *GFP* were mixed with *A. tumefaciens* carrying *E4A*, *E4B*, *E4C* and *INF1* respectively at final optical density at 600 nm (OD_{600}) of 0.2, then infiltrated in each half of *Nicotiana benthamiana* leaves. (a) and (d) suppression of cell death by *E4D* compared with *GFP* on typical leaves. The leaves were decolorized by ethanol to show the remarkable difference. The white dotted circle indicates the infiltrated areas without cell death and the black dotted circle indicates the cell death areas. (b) and (e) statistical analysis of cell death on *N. benthamiana* leaves scored at 4 dpi. Asterisks represent significant differences from *GFP* control (Wilcoxon rank-sum test; *** $p < 0.001$). (c) Protein accumulation was detected by anti-Flag antibodies and anti-mCherry antibodies by western blot. *E4D* was fused with *mCherry*, and the other proteins were fused with 3*Flag. Protein loading is indicated by Ponceau S stain (PS). Three independent experiments were conducted with the same results.

result demonstrated that there is no direct interaction between *E4A*-GFP and Flag-*E4D*, implying that the inhibitory effect of *E4D* on other alleles is not due to direct interaction between them (Figure S4). We speculate that this inhibition occurs via competition for a common host target.

Truncated *PpE4* mutants deficient in cell death-inducing activity showed abolished ability to increase plant susceptibility

To validate that the virulence function of *PpE4* is related to its cell death, we further investigated the contribution of a series of *PpE4* truncated mutants to

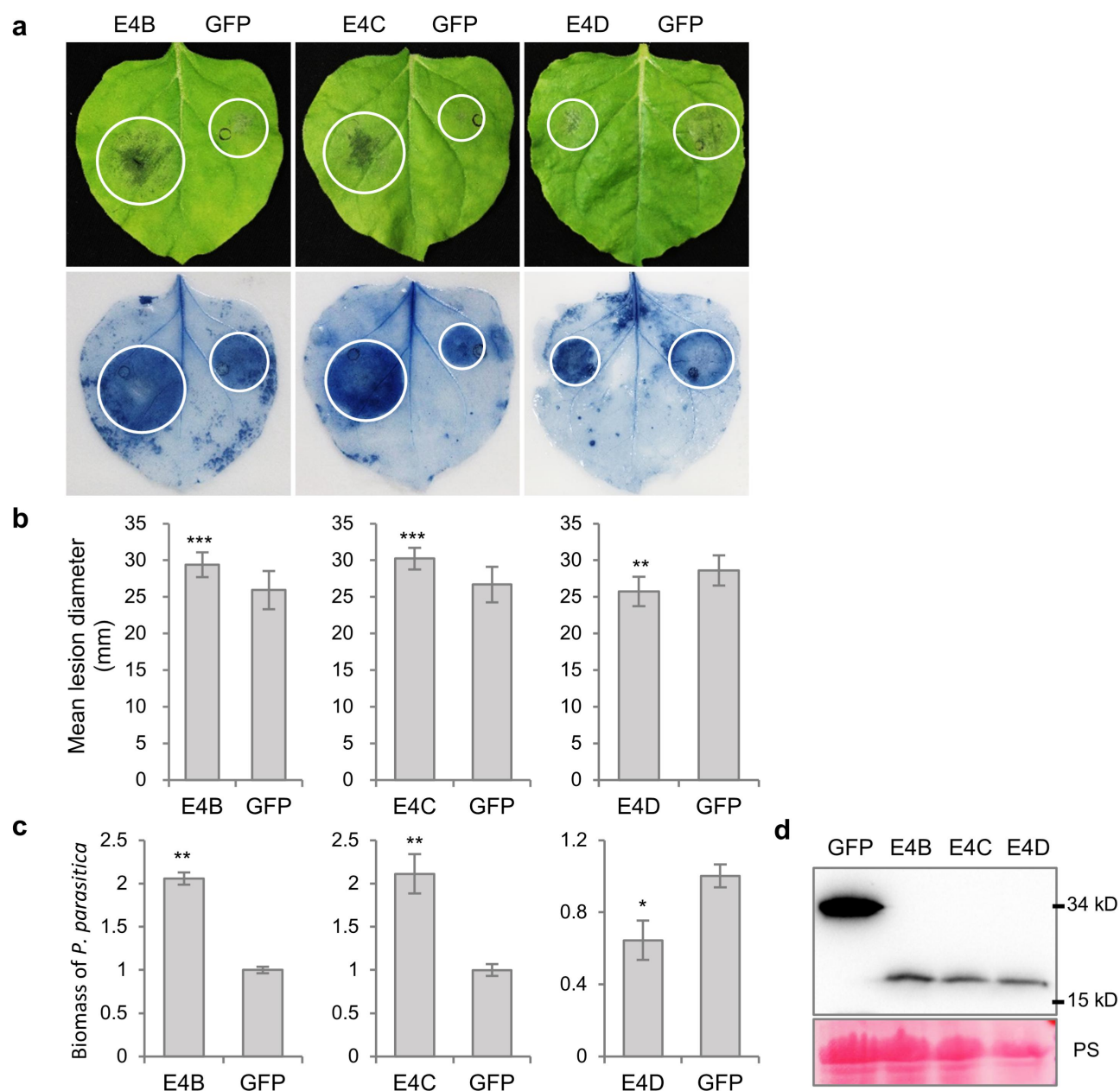


Figure 5. The contributions of *PpE4* alleles to *Phytophthora parasitica* infection. (a) Typical water-soaked lesions formed on leaves. Trypan blue stain was used to highlight the infection area. (b) Mean lesion diameter on *Nicotiana benthamiana* leaves measured at 48 hours post-inoculation. Error bars represent the standard deviation (SD) of at least 9 leaves. Asterisks represent significant differences from *GFP* control (two tailed *t*-test; ** $p < 0.01$; *** $p < 0.001$). (c) Biomass of *P. parasitica* in *N. benthamiana* leaves quantified by qPCR. Bars represent relative ratio of *PpUBC* DNA to *NbF_box* DNA with SD of three biological replicates. Asterisks represent significant differences from *GFP* control (two tailed *t*-test; * $p < 0.05$; ** $p < 0.01$). (d) Proteins accumulation was detected by anti-Flag antibodies using western blot. Protein loading is indicated by Ponceau S stain (PS). Three independent experiments were conducted with similar results.

P. parasitica infection upon their transient expression in *N. benthamiana*. According to a previous study, the mutants E4M1, E4M4, and E4M8 were able to trigger cell death, while E4M3 and E4M5 were unable to do so [36]. Consistent with our hypothesis, the mutants that were capable of triggering cell death were also able to promote infection,

whereas the other two mutants that lacked this ability were unable to promote infection (Figure 7). The results showed that residues 56–121 of *PpE4* play a dual role in triggering cell death and promoting infection, suggesting that *PpE4*'s contribution to pathogen infection is linked to its cell death-inducing activity.

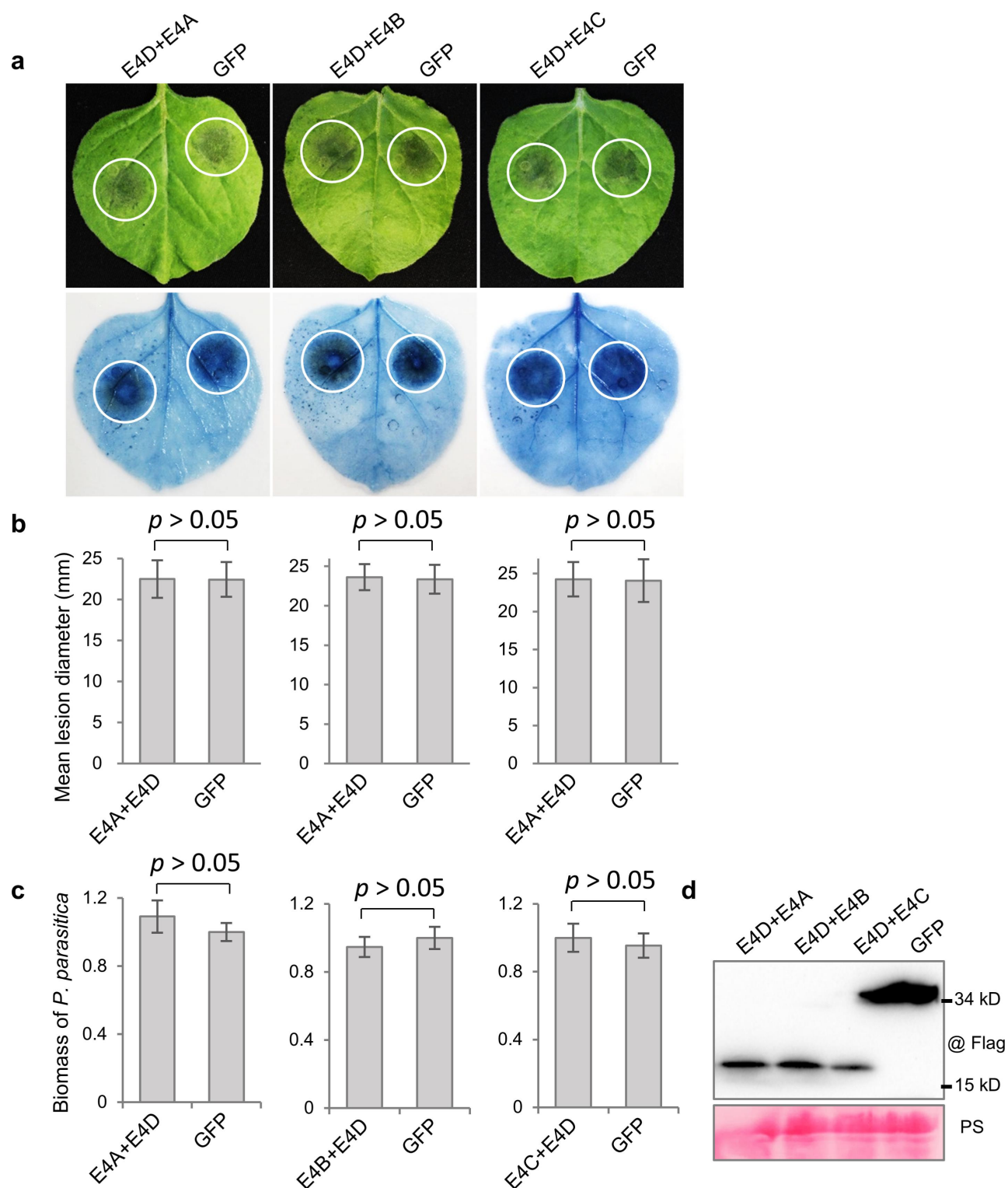


Figure 6. *E4D* suppresses the virulence mediated by the other three alleles in *planta*. *agrobacterium tumefaciens* carrying Flag-*E4D* was co-injected with *A. tumefaciens* carrying Flag-*E4A*, Flag-*E4B*, and Flag-*E4C* in each half of *Nicotiana benthamiana* leaves at final optical density at 600 nm (OD_{600}) of 0.2 followed by inoculation with *Phytophthora parasitica*, GFP was expressed in the other half as control. (a) Typical water-soaked lesions formed on leaves. Trypan blue stain was used to highlight the infection area. (b) Mean lesion diameter on *N. benthamiana* leaves measured at 48 hours post-inoculation. Bars represent mean lesion diameter with standard deviation (sd) from at least 15 biological replicates (two tailed *t*-test; $p > 0.05$). (c) Biomass of *Phytophthora parasitica* in *N. benthamiana* leaves quantified by qPCR. Bars represent relative ratio of *PpUBC* DNA to *NbF_box* DNA with sd of three biological replicates (two tailed *t*-test; $p > 0.05$). (d) Proteins accumulation was detected by anti-Flag antibodies using western blot. Protein loading is indicated by Ponceau S stain (PS). Three independent experiments were conducted with similar results.

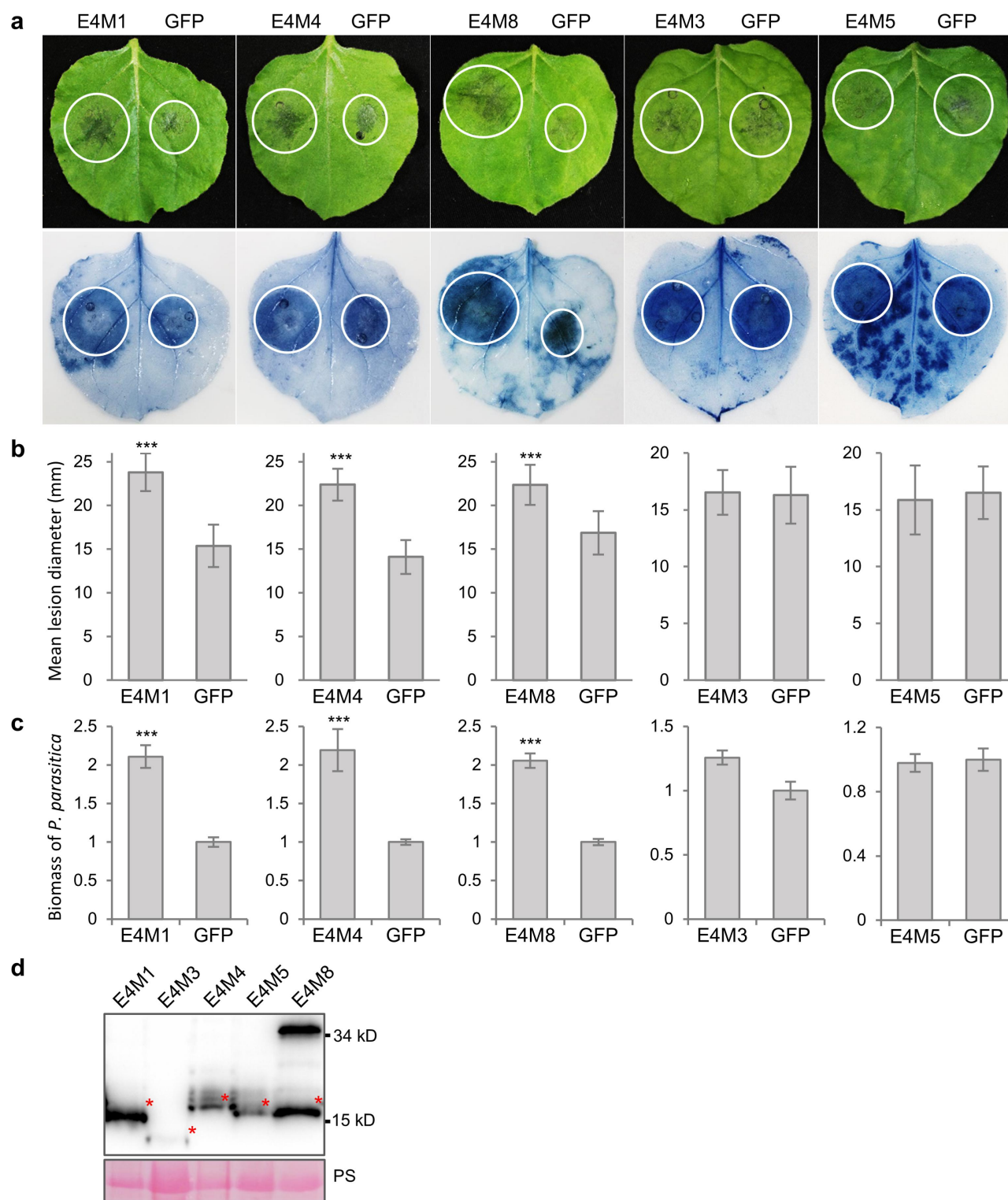


Figure 7. Truncated *PpE4* mutants without cell death-inducing activity failed to promote infection. The *PpE4* truncated mutants were transiently expressed on one side of *Nicotiana benthamiana* leaves followed by inoculation with *Phytophthora parasitica*, GFP was used as a control in the other side ($OD_{600} = 0.01$). (a) Photographs of typical leaves were taken at 48 hours post-inoculation (hpi), and stained by trypan blue. (b) Mean lesion diameter on *N. benthamiana* leaves were measured at 48 hpi. Bars represent mean lesion diameter with standard deviation (sd) from at least 16 biological replicates. Asterisks represent significant differences from GFP control (two tailed *t*-test; *** $p < 0.001$). (c) Biomass of *P. parasitica* in *N. benthamiana* leaves quantified by qPCR. Bars represent relative ratio of *PpUBC* DNA to *NbF_box* DNA with sd of three biological replicates. Asterisks represent significant differences from GFP control (two tailed *t*-test; *** $p < 0.001$). (d) Protein accumulation of *PpE4* truncated mutants detected by anti-Flag antibody using western blot. Protein loading is indicated by Ponceau S stain (PS). Three independent experiments were conducted with similar results.

Cell death is required for *PpE4* to increase plant susceptibility

According to previous study, *PpE4*-induced cell death was almost abolished in *NPK*-silenced *N. benthamiana* plants [36]. To further confirm that the cell death of *PpE4* is necessary for its virulence, *P. parasitica* was inoculated following the transient expression of *PpE4* and *GFP* on *NPK*-silenced or *GFP*-silenced *N. benthamiana* leaves. The results showed that *PpE4* expression enhanced colonization by pathogen on the *GFP*-silenced plants, but had no impact on the colonization by pathogen on *NPK*-silenced plants (Figure 8). Thus, it can be concluded that *PpE4* failed to promote infection when its cell death was blocked on the *NPK*-silenced plants. In other words, the cell death activity is required for *PpE4* to increase plant susceptibility and promote pathogen infection.

Discussion

It has been reported that the RXLR effectors with important virulence functions and cell death-inducing activity are often polymorphic in pathogen populations due to positive selection pressure. The most conspicuous example is the *Avr* genes, which exhibit frequent copy number variations, presence or absence polymorphism, coding sequences mutations and gene expression differentiation [24,43–45]. A previous study revealed that *PpE4* is highly expressed during early infection stages and possesses dual activities, facilitating infection and triggering plant cell death [36]. Therefore, *PpE4* is supposed to be subjected to positive selection pressure. To validate this hypothesis, a polymorphism assay was carried out to show that *PpE4* displays a high level of sequence variation in the *P. parasitica* population, including point mutations and presence/absence polymorphisms (Figure 1). With a dN/dS ratio of 1.14 at the C-terminal “effector” domain, it is

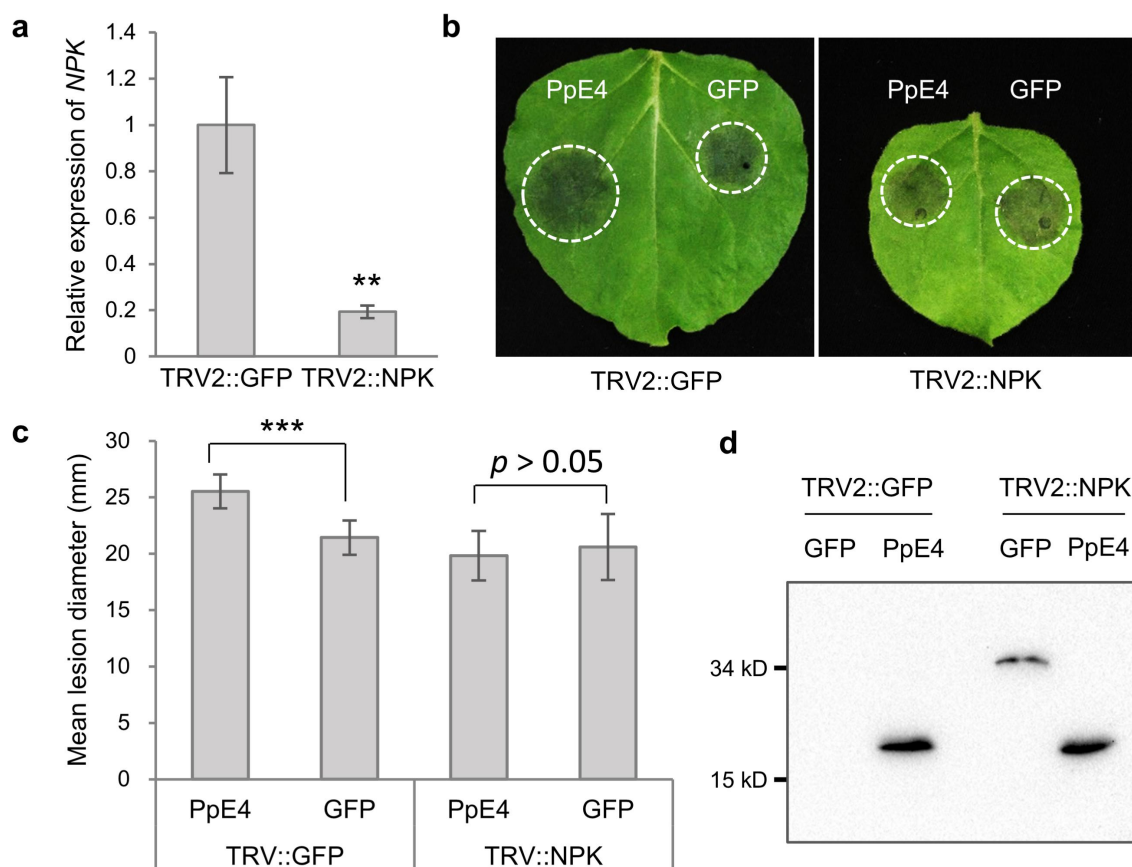


Figure 8. *PpE4* failed to promote infection on *NPK*-silenced *Nicotiana benthamiana*. *Agrobacterium tumefaciens* carrying *PpE4* and *GFP* were infiltrated in each side of *NPK*-silenced or *GFP*-silenced *N. benthamiana* leaves one day before inoculation of *Phytophthora parasitica* 1121. (a) *NPK* expression level was down-regulated by 80% in TRV2:*NPK* plants as compared to the TRV2:*GFP* plants at 20 days after infiltration. The constitutively expressed β -actin gene was used as a reference gene for RT-qPCR assay. (b) Typical water-soaked lesions formed on *NPK*-silenced or *GFP*-silenced leaves taken at 48 hours post-inoculation (hpi). (c) Mean lesion diameter measured at 48 hpi. Asterisks represent significant differences from *GFP* control (two tailed *t*-test; ****p* < 0.001). (d) Western blot detection of *PpE4* and *GFP* accumulation in vigs plants by anti-Flag antibody. Three independent experiments were conducted with similar results.

evident that *PpE4* was evolving rapidly in response to positive selection pressure during the co-evolutionary arms race between the host and pathogen. Out of the four alleles, three are still capable of triggering cell death, whereas the substitution of six amino acids in the 56–121 region of E4D allele led to the loss of its cell death-inducing activity. Through a combination of natural variation and point mutagenesis assays, we identified that the F74L mutation deprives E4D of its ability to trigger cell death. It was reported that *PaRXLR24*, *Avh238*, and *PpE4* are orthologs [42]. The equivalent site of *PpE4* F⁷⁴ in *Avh238* is “F,” and “Y” in *PaRXLR24*. Moreover, the critical position for the cell death-inducing activity of *Avh238* is H⁷⁹, while the equivalent positions in *PaRXLR24* and the four *PpE4* alleles are I⁸¹ (Figure S5). This suggests that these two amino acid sites may be critical for the cell death-inducing activity of these three orthologs [25,42]. Apart from point mutations, *PpE4* has also been found to be deleted from the genome of several isolates, as confirmed by RT-qPCR, PCR, and genome sequencing data (Figure 1). This absence represents a new DNA polymorphism in the *PpE4* locus, similar to *PiAVR2*, *PsAvr1b-1*, and *PsAvr1d*, which also exhibit presence/absence polymorphisms in addition to sequence variation [24,43,44]. As reported, the presence or expression of a *PiAVR2*-like gene functionally complements the silencing or absence of *PiAVR2* [44]. Due to the fast evolution and redundancy of various RXLR genes, it is possible that pathogens have developed new effector genes to replace *PpE4* in the *P. parasitica* genome in the absence of it.

In this study, we found that the *E4D* allele which lost its cell death-inducing activity, also lost its ability to promote infection (Figure 1 and 5). Unlike *Avh238*, which underwent a mutation in a single amino acid to escape recognition, yet still retained its immunosuppressive activity in plants to enhance pathogen virulence [25]. Additionally, the *E4D* allele lacking cell death-inducing activity could suppress the cell death triggered by the other three alleles as well as neutralize their virulence function (Figure 4 and 6). This scenario mirrors the dominant-negative effect of the kinase-inactive mutant on CRN8, which inhibits CRN8-induced cell death and virulence [46]. We speculate that the inhibitory effect of E4D on other alleles occurs through competition for a common host target, since E4D did not interact with E4A. Alternatively, there is another explanation that proposes a potential correlation between the virulence of *PpE4* and its capacity to induce cell death. Two experiments were conducted to test this hypothesis. First, the virulence function of the *PpE4* truncated mutants with or without cell death-inducing activity was investigated after being expressed

in *N. benthamiana*. The results showed that infection was promoted by three mutants (E4M1, E4M4, and E4M8) that trigger cell death, whereas the other two mutants (E4M3 and E4M5), which lack cell death-inducing activity, did not promote infection (Figure 7). This suggests that the 56–121 residues of *PpE4* are responsible for both triggering cell death and promoting infection. Secondly, we investigated whether the expression of *PpE4* could promote pathogen infection even when its cell death was blocked in NPK-silenced plants. As expected, *PpE4* failed to promote infection when its cell death was blocked (Figure 8), thus reinforcing the idea that the cell death-inducing activity of *PpE4* was not contradictory to its virulence, but rather necessary for its virulence function of enhance plant susceptibility. However, not all cell death triggered by effectors is associated with their virulence function. For example, the cell death induced by *PcCRN83_152* is not correlated with its virulence function [47], and *Avh238* only contributes to pathogen virulence when its cell death is suppressed [25]. *E4D* was able to delay *INF1*-induced cell death, suggesting that *E4D* may interfere with the signaling pathway downstream of *INF1*- and *PpE4*-induced cell death. This investigation will illuminate the specificity of *E4D*'s inhibitory effect and its broader role in regulating plant immunity.

In most situations, plant cell death triggered by RXLR effectors is considered a form of disease resistance. This is because the recognition of the pathogen's avirulence effector by the host NLR receptor results in the hypersensitive response (HR), which is a type of programmed cell death regulated by the plant to limit the spread of the pathogen at the cost of local cell death. Given that the majority of avirulence proteins encoded by oomycete pathogens belong to the RXLR effectors [7], it can be inferred that if an RXLR effector triggers cell death in the host plant, it is highly likely to trigger an NLR-mediated plant immune response. For instance, the cell death induced by *PaRXLR24*, an ortholog of *PpE4* from the kauri dieback pathogen *P. agathidicida*, was found to be mediated by potential NLR receptors in *N. benthamiana* [42]. Nevertheless, numerous RXLR effectors have been found to trigger cell death in plants, most of which lack identified recognition receptors [26,28–30]. In contrast to the specific HR occurred in the gene-for-gene model, the cell death triggered by *PpE4* is nonspecific and broad-spectrum [36]. This suggests that the recognition receptor is potentially highly conserved across different plant species. An alternative hypothesis suggests that cell death is caused by effectors hijacking essential host components to support the hemibiotrophic life

cycle, such as *P. capsica* CRISIS2 [32] and *P. infestans* PITG06478 [33]. Moreover, necrotrophic fungal pathogens take advantage of plant cell death to promote infection by secreting victorin, which elicits an NLR-mediated cell death [35]. This indicates that the proper timing and manner of cell death is beneficial for the successful infection of both hemibiotrophic and necrotrophic pathogens [6]. *P. parasitica* is a pathogen with a wide host range, as almost all tobacco and *Arabidopsis* are susceptible to it, and no race-specific resistance has been found in either tobacco or *Arabidopsis*. This may explain why *PpE4* triggers cell death during *P. parasitica* infection, while the pathogen's colonization in various hosts is not restricted. Despite the fact that RXLR effectors such as *PpE4* can induce cell death, this particular type and timing of cell death does not limit pathogen colonization in the conventional sense. Instead, it was controlled by the *P. parasitica* to facilitate the expansion of the pathogen.

In conclusion, we found that *PpE4* is polymorphic in the *P. parasitica* population and identified an allele that lacks both cell death-inducing ability and virulence function. Instead, it can counteract the cell death-inducing and virulence functions of the other three alleles. Further investigation showed that truncated mutants of *PpE4* without cell death-inducing activity were incapable of promoting infection. Additionally, *PpE4* was ineffective in facilitating infection in *NPK*-silenced plants, where its cell death was blocked. In summary, we concluded that the cell death-inducing activity of *PpE4* is required for its contribution to hemibiotrophic pathogen colonization, suggesting that *P. parasitica* exploits the cell death induced by *PpE4* to facilitate the transition from biotrophic to necrotrophic phase. Our findings provide new insights into the biological function of cell death induced by RXLR effectors, and lay a foundation for understanding the pathogenicity of hemibiotrophic pathogens.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Author contributions

CRedit: **Guiyan Huang:** Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing; **Yanan Hu:** Investigation; **Nannan Lu:** Investigation; **Yuling Meng:** Investigation, Methodology; **Weixing Shan:** Conceptualization, Funding acquisition, Resources, Supervision, Visualization, Writing – review & editing.

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Data availability statement

The source data of this study are openly available at <https://doi.org/10.5281/zenodo.17226757> [48].

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